

2(a)(i)	any two from: <i>idea of synthesis from smaller units</i> 1 DNA replication / transcription ; 2 to, make / build up, (named) protein ; 3 to, make / build up, (named), lipid / triglyceride / phospholipid ; 4 to, make / build up, (named) carbohydrate / polysaccharide ; 5 phosphorylation ;	2
2(a)(ii)	any three from: 1 phosphorylates glucose or glucose → glucose / fructose / hexose, phosphate ; 2 second phosphorylation or glucose / fructose / hexose, phosphate → fructose / hexose, bi(s)phosphate ; 3 so glucose cannot leave the cell ; 4 activates / increases energy of / decreases stability of, glucose ;	3
2(b)(i)	any three from: <i>decreased production of ATP</i> 1 smaller / less steep, proton gradient ; 2 few(er) protons go through, ATP synth(et)ase / stalked particle ; <i>increased lipid metabolism</i> 3 use / burn / respire, body fat / lipid stores ; 4 as less ATP made from, glucose / food ;	3
2(b)(ii)	any two from: 1 more / faster / increased, (rate of) glycolysis ; 2 oxygen, decreased / used faster / shortage ; 3 (so) more / increased, <u>anaerobic</u> respiration ;	2
2(b)(iii)	any one from: energy from, reduced NAD / ETC, released as heat ; energy not used to join ADP and Pi so released as heat ;	1